

Randomized Phase III Evaluation of Cisplatin Plus Fluorouracil Versus Cisplatin Plus Paclitaxel in Advanced Head and Neck Cancer (E1395): An Intergroup Trial of the Eastern Cooperative Oncology Group

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The Southwest Oncology Group also participated in the study.

Authors' disclosures of potential conflicts of interest are found at the end of this article.

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ABSTRACT

Purpose

To determine the response rate, survival and toxicity of infusional cisplatin plus fluorouracil (CF) versus cisplatin plus paclitaxel (CP) in patients with incurable squamous cell cancer of the head and neck, with the hypothesis that CP is superior.

Patients and Methods

Two hundred eighteen patients with locally advanced, recurrent, or metastatic disease were randomly assigned to CF (cisplatin 100 mg/m² day 1 and fluorouracil 1,000 mg/m²/24 hours by continuous intravenous infusion day 1 through 4) or CP (cisplatin 75 mg/m² day 1 and paclitaxel 175 mg/m² over 3 hours on day 1). Cycles were repeated every 3 weeks until progression or a minimum of 6 cycles with complete response or stable disease. The primary outcome was overall survival. Secondary outcomes included response rate and toxicity.

Results

No significant difference in overall survival or response rate was seen. Estimated median survival was 8.7 months in the CF group and 8.1 month in the CP group. Objective response rate (complete response plus partial response) was 27% in the CF group and 26% in the CP group. Toxicity was similar between groups, with the most frequent including myelosuppression, thrombocytopenia, anemia, nausea, vomiting, and stomatitis. A total of 12 deaths occurred (CF, seven; CP, five) during treatment; eight from infection, two from hemorrhage, one from cardiac causes and one from unknown causes. Gastrointestinal and hematologic toxicities were more common in the CF group, whereas neurotoxicity was equivalent between groups.

Conclusion

This phase III, randomized, multicenter trial showed no difference in survival between patients treated with CF or CP.

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INTRODUCTION

Recurrent or metastatic squamous cell cancer of the head and neck is a common problem that remains difficult to treat, due to both intrinsic characteristics of the tumor as well as the health status of this patient population. Early work in this tumor type investigated the activity of single-agent methotrexate, while

subsequent studies utilized combinations of agents that included fluorouracil (FU), cisplatin, and alkylators. Common to these single- and multiagent studies was the finding of better response rates (RRs) with the combinations but no improvement in survival compared with methotrexate alone.¹⁻⁴

With the development of the taxanes in the 1990s came the opportunity to conduct

a series of clinical trials to investigate the activity in head and neck cancer of this novel cytotoxic agent. The initial study⁵ by the Eastern Cooperative Oncology Group (ECOG) utilized the recommended single-agent dose and schedule of 250 mg/m² by 24-hour infusion administered every 3 weeks. This phase II study of 34 patients demonstrated a 40% RR with a median survival of 9.2 months and a 1-year survival of 32%. Although granulocytopenia was common, the results suggested improved activity compared with historical controls treated with other single agents.

On the basis of the encouraging results seen with paclitaxel given alone, ECOG then initiated a phase III trial to investigate the activity of two combinations of this drug with cisplatin. Both regimens utilized the 24-hour infusion schedule of paclitaxel (either 175 mg/m² or 200 mg/m²) administered with cisplatin (75 mg/m²) and growth factor support.⁶ There was no difference in RR or survival; however, severe hematologic toxicity contributed to early treatment termination in many patients treated with both doses.

Subsequent phase II trials^{7,8} that used a 3-hour infusion of paclitaxel in combination with cisplatin revealed similar activity with much less hematologic and overall toxicity, thus providing a safer regimen with equal efficacy for use in future trials. While these trials yielded a safe and active paclitaxel-based combination regimen, it remained unclear whether this doublet was better than standard approaches.

As such, in 1997, ECOG initiated a randomized phase III trial that compared cisplatin and paclitaxel with the combination considered the standard—cisplatin and infusional FU. The Southwest Oncology Group (SWOG) joined the study in February of 1999. Previous multicenter trials of cisplatin and FU found RRs of approximately 30%, median survival times of 4 to 6 months, and 1-year survival rates of 20% to 25%. The goal of this trial was to compare the efficacy and toxicity of the two regimens, cisplatin plus FU (CF) and cisplatin plus paclitaxel (CP). The primary end point was overall survival at 1 year. Secondary end points included response and toxicity.

PATIENTS AND METHODS

Eligibility criteria

All patients had either measurable or assessable, histologically confirmed, squamous cell carcinoma of the head and neck (excluding nasopharyngeal carcinoma), which was not curable with surgery or radiation therapy. Two patient groups were eligible: those with previously untreated extensive local-regional disease or distant metastases, and those with previously treated disease and subsequent local-regional recurrence, persistent disease, or distant metastases. Measurable disease in a previously irradiated area was acceptable if progressive or biopsy proven. Prior chemotherapy for recurrent disease was not allowed. Prior chemotherapy delivered as part of initial curative therapy was allowed; however, treatment with paclitaxel or FU had to be com-

pleted more than 12 months before study entry and treatment with cisplatin had to be completed more than 6 months before study entry. Additional requirements included: age $\geq$ 18 years; ECOG performance status (PS) of 0 or 1; adequate bone marrow reserve (defined as absolute neutrophil count $\geq$ 1,500/mm³, hemoglobin $\geq$ 10 gm/dL, platelets $\geq$ 100,000/mm³); creatinine clearance $\geq$ 50 mL/min; total bilirubin $\leq$ 1.5 mg/100 mL; no concurrent malignancy; no significant cardiac disease within the previous year; no allergy to Cremophor or *E coli*-derived proteins; and no brain metastases. Written informed consent in compliance with institutional and federal guidelines was required, and the protocol was approved by the institutional review board of each participating institution.

Randomization and Treatment

Eligible patients were centrally randomly assigned to CF or CP treatment groups by using permuted blocks within strata with dynamic balancing within main institutions and their affiliated networks. They were also stratified by disease status (newly diagnosed v previously treated) and PS (ECOG 0 v 1).

Treatment with CF (cisplatin 100 mg/m² day 1 and FU 1,000 mg/m²/24 hours by continuous intravenous infusion days 1 through 4) and CP (cisplatin 75 mg/m² day 1 and paclitaxel 175 mg/m² over 3 hours on day 1) was administered every 3 weeks. Carboplatin (area under the curve [AUC], 6) was substituted for cisplatin in both treatment groups for development of grade 2 or greater neuropathy or renal impairment (creatinine clearance $<$ 50 mL/min). Cisplatin was administered with hydration and forced diuresis, and patients underwent routine monitoring of electrolytes, serum creatinine, and magnesium. Patients receiving paclitaxel were premedicated with 20 mg of oral dexamethasone at 12 and 6 hours before the infusion and with intravenous diphenhydramine and an H2 receptor antagonist within 60 minutes before the infusion of paclitaxel to reduce the risk of hypersensitivity reaction.

Dose delays or modifications were specified for hematologic, gastrointestinal, neurologic and renal toxicities. For all grade 4 toxicities, treatment was withheld until complete resolution of the toxicity and then resumed with dose modification. Hematologic requirements for subsequent cycles included an absolute neutrophil count $\geq$ 1,500/mm³ and a platelet count $>$ 100,000/mm³ on day 1 of treatment. Treatment was delayed until recovery to these values. All chemotherapy doses were reduced by 20% for nadir grade 4 thrombocytopenia or nadir grade 4 neutropenia persisting longer than 5 days or for febrile neutropenia requiring hospitalization and antibiotics. Grade 3 mucositis required a 20% dose reduction of FU and paclitaxel in subsequent treatment cycles.

For grade 2 hepatic toxicity, treatment was held until toxicity resolved to grade 1 and then administered without dose reduction. For grade 3 hepatic toxicity, paclitaxel was withheld until toxicity resolved to grade 1 and then administered at 80% of the previously calculated dose. For grade 2 sensory or motor neuropathy, carboplatin was substituted for cisplatin, and the doses of FU and paclitaxel were not changed. If grade 2 neuropathy persisted for an additional cycle, then paclitaxel was reduced to 80% of the previously calculated dose. For grade 3 neuropathy, patients were removed from study at the discretion of the treating physician. For a creatinine clearance of $\leq$ 50 mL/min, carboplatin was substituted for cisplatin.

Response was assessed after every two treatment cycles. Patients achieving a complete response (CR) could discontinue treatment after a minimum of six cycles that included at least two cycles after documentation of the CR. Patients achieving a partial

response (PR) continued treatment until evidence of CR or progression of disease (PD). Patients with stable disease (SD) could discontinue treatment after six cycles.

Response Criteria

Both measurable and assessable disease were allowed in this study. Measurable disease could be bidimensional or unidimensional. Nonmeasurable, assessable disease consisted of malignant disease evident on clinical (physical or radiographic) examination but not measurable by ruler or calipers. Examples of this include pelvic or abdominal masses; lymphangitic or confluent multinodular lung metastases; ascites or pleural effusions; and disease noted on liver scans, bone scans, or gallium scans. For nonmeasurable, assessable disease, a PR was defined as definite improvement estimated to be in excess of 50% (agreed on by two independent investigators) for at least 4 weeks. A partial response of bony metastases was defined as a partial decrease in the size of the lesions, blastic transformation of lytic lesions or decreased density of blastic lesions, lasting for at least 4 weeks.

Standard ECOG solid tumor response criteria for measurable disease were used. CR was defined as complete disappearance of all clinically detectable malignant disease for at least 4 weeks. PR was defined as at least a 50% decrease in tumor size for at least 4 weeks without an increase in size of any area of known malignant disease of greater than 25% or appearance of new areas of malignant disease. SD was defined as no significant change (decrease of < 50% or increase of < 25%) in measurable disease for at least 4 weeks and no new areas of malignant disease. PD was defined as at least a 25% increase in size of any malignant lesion or the appearance of new malignant lesions.

Study Design and Statistical Methods

This is a randomized, phase III study comparing two treatment regimens—CF and CP—with the hypothesis that the CP combination is superior. The sample size calculation was based on the major end point of overall survival. The study was designed to detect a 15% difference in 1-year survival (20% v 35%) between the two treatment regimens with 80% power and a type I error of .05 using a two-sided log-rank test and an exponential survival model. This model required accrual of 212 patients over 2.25 years.

Secondary end points included RRs, toxicity, quality of life, and pain intensity. The quality of life assessment was determined using the Functional Assessment of Cancer Therapy for Head and Neck Cancer (FACT-H&N). Patients completed the evaluation before chemotherapy, at baseline, and at week 7, week 16, and month 6. At similar time points, patients also completed the Brief Pain Inventory.⁹

The trial was monitored for early stopping according to the principles of group-sequential methods using the O'Brien-Fleming boundary.¹⁰ Analyses were scheduled when approximately 87, 130, and 176 deaths occurred or when 49%, 74%, and 100% of the information required for 80% power was available. These calculations assumed constant rates of mortality.

Fisher's exact test was used to analyze the contingency tables of response. The Kruskal-Wallis test was also performed to compare the distribution of categorical data across the treatment arms. Survival data were analyzed using the Kaplan-Meier method, and comparison between groups was done using the log-rank test.¹¹ Cox proportional hazards regression was used to model the survival data based on the covariates of interest.¹² Unless specified, the *P* values reported are for two-sided tests.

RESULTS

Patient Characteristics

A total of 218 patients, with 109 in each group, entered the trial. The study was opened in March 1997 and closed in January 2000 after accrual goals were met. Of the 218 registered patients, 20 (CF, seven; CP, 13) were ineligible and two (both CF) did not receive assigned treatment. Patients were ineligible for the following reasons: no baseline ALT (CF, two; CP, four), no baseline PS (CF, one; CP, six), resectable disease (CF, one), unknown primary (CF, one), skin cancer primary (CP, one), history of recent myocardial infarction (CP, one), no established head and neck cancer (CF, one), no baseline measures (CF, one), and baseline hemoglobin of 9.8 (CP, one). The six excluded patients with no baseline ALT were included in the analysis, yielding 204 patients for the intention to treat analysis.

Baseline characteristics of the remaining 204 patients are listed in Table 1. The main demographic features of age, sex, and race were similar between groups. The mean age was 61 years, and most patients were male. As expected with

Table 1. Baseline Patient Characteristics

Characteristic	CF Arm (n = 104)		CP Arm (n = 100)		Total
	No.	%	No.	%	
Sex					
Male	87	83.6	78	78.0	165
Female	17	16.4	22	22.0	39
Race					
White	83	79.8	77	77.0	160
Non-white	21	20.2	23	23.0	44
Age, years					
Range	35-84		37-81		35-84
Median	61		61		61
Performance status					
0	29	27.9	25	25.0	54
1	74	71.1	75	75.0	149
2	1	1.0	0	0.0	1
Disease status					
Newly diagnosed	14	13.5	11	11.0	25
Recurrent	90	86.5	89	89.0	179
Metastases					
No	29	27.9	38	38.0	67
Yes	63	60.6	52	52.0	115
Unknown	12	11.5	10	10.0	22
Primary site					
Oral cavity	18	17.3	26	26.0	44
Oropharynx	28	26.9	23	23.0	51
Hypopharynx	18	17.3	9	9.0	27
Larynx	26	25.0	26	26.0	52
Other*	5	4.8	5	5.0	10
> 1 site	3	2.9	7	7.0	10
None identified	6	5.8	4	4.0	10

Abbreviations: CF, cisplatin plus fluorouracil; CP, cisplatin plus paclitaxel.
*Includes salivary glands, lip, paranasal sinus, middle ear, and nose.

this population of patients with advanced or recurrent disease, most patients had a PS of 1 (CF, 71.1%; CP, 75.0%) with recurrent (CF, 86.5%; CP, 89.0%) and metastatic (CF, 60.6%; CP, 52.0%) disease. There were small differences in the distribution of tumor site between groups, with more hypopharyngeal tumors in the CF group and more oral cavity tumors in the CP group. Overall however, the groups were well matched for baseline characteristics.

Toxicity

Toxicity information is reported for all patients who received any treatment (two assigned patients were not treated). A total of 12 deaths occurred (CF, seven; CP, five) during treatment: eight from infection, two from hemorrhage, one from cardiac causes, and one from unknown causes. By definition, patients in the CF arm received a higher dose of cisplatin (100 mg/m² v 75 mg/m²) than those in the CP arm. The percentages of grade 3, 4, and 5 toxicities are listed in Table 2. Hematologic adverse effects occurred most often, were more common in the CF group, and consisted mainly of myelosuppression and infection followed less frequently by anemia and thrombocytopenia.

Nausea and vomiting were equally common, whereas diarrhea was slightly more frequent in the CF group. As expected, stomatitis was more frequent in the group that received infusional FU (CF, 31% grade 3 and 4; CP, 0). Neurotoxicity and metabolic events, however, were equivalent between the groups. A total of 28 patients switched

from cisplatin to carboplatin due to development of the following toxicities: neurologic (CF, four; CP, two), renal (CF, 13; CP, four), other (CF, four), and unknown (CP, one). The Kruskal-Wallis test for comparing the distribution of worst-degree (grade 3 or higher) toxicities between the two treatments indicated no significant difference between the groups.

Response and Survival

Overall survival was not different between the two groups. The 1-year survival rates (CF, 41.4%; CP, 32.4%; log-rank $P = .49$) and total number of deaths (CF, 95 of 104; CP, 90 of 100) did not differ between groups. With an overall median follow-up of 8.3 months, estimated median survival was 8.7 months (95% CI, 6.7 to 12.2) in CF, and 8.1 months (95% CI, 6.1 to 10.0) in CP.

Evaluation of RR was a secondary end point of the trial. While most patients had measurable disease, a total of 17 patients had assessable disease only. A listing of best response by treatment group is provided in Table 3. In 25 cases (CF, 11; CP, 14) response was not available, mostly due to lack of repeated measurements. For the 204 patients included in the analysis, 27.9% reached a CR or PR (CF, 29.8%; CP, 26.0% in). Stable disease was achieved in most of the remaining patients, but this was not durable. Overall, there was no difference in RRs between treatment groups ($\chi^2 P = .84$).

The Kaplan-Meier curve for overall survival is presented in Figure 1. No survival difference is evident by the log-rank test ($P = .49$). Multiple Cox proportional hazards regression models were used to evaluate for the impact of prognostic factors on survival. Only PS was predictive of survival. In a multivariate model controlled for treatment group, age, sex, race, disease status, metastasis, and histologic differentiation, the hazard ratio for death comparing patients with PS ≥ 1 to those with PS = 0 was 1.47 (95% CI, 1.02 to 2.12), indicating poorer survival in patients with lower PS.

Table 2. Grade 3 to 5 Toxicities (by arm)

Toxicity	CF Arm (n = 106)			CP Arm (n = 108)		
	3 (%)	4 (%)	5 (%)	3 (%)	4 (%)	5 (%)
Leukopenia	45	18	—	27	8	—
Granulocytopenia	27	40	—	25	30	—
Thrombocytopenia	17	6	—	3	1	—
Anemia	31	2	—	9	4	—
Infection	15	2	4	8	1	4
Genitourinary	2	1	—	—	1	—
Nausea	18	1	—	17	1	—
Vomiting	11	7	—	6	4	—
Diarrhea	3	3	—	1	—	—
Stomatitis	24	7	—	—	—	—
Hemorrhage	—	1	1	—	—	1
Mucositis	1	—	—	—	—	—
Liver	1	—	—	3	—	—
Cardiac	2	—	1	2	2	—
Hypotension	2	—	—	3	2	—
Neurosensory	4	—	—	5	—	—
Neuromotor	3	—	—	4	—	—
Metabolic	11	4	—	8	2	—
Fatigue	8	1	—	4	3	—
Dehydration	3	2	—	4	—	—
Other	5	—	1	5	1	—
Worst Overall	40	50	7	33	42	5

Abbreviations: CF, cisplatin plus fluorouracil; CP, cisplatin plus paclitaxel.

Table 3. Best Response by Treatment Arm

Response	CF Arm		CP Arm	
	No. of Patients	%	No. of Patients	%
Complete	7	6.7	7	7.0
Partial	24	23.1	19	19.0
Stable	58	55.8	54	54.0
Progression	4	3.8	6	6.0
Not assessable	11	10.6	14	14.0
Total	104	100.0	100	100.0

NOTE. $\chi^2 P = .84$. Seventeen patients had assessable disease only. Abbreviations: CF, cisplatin plus fluorouracil; CP, cisplatin plus paclitaxel.

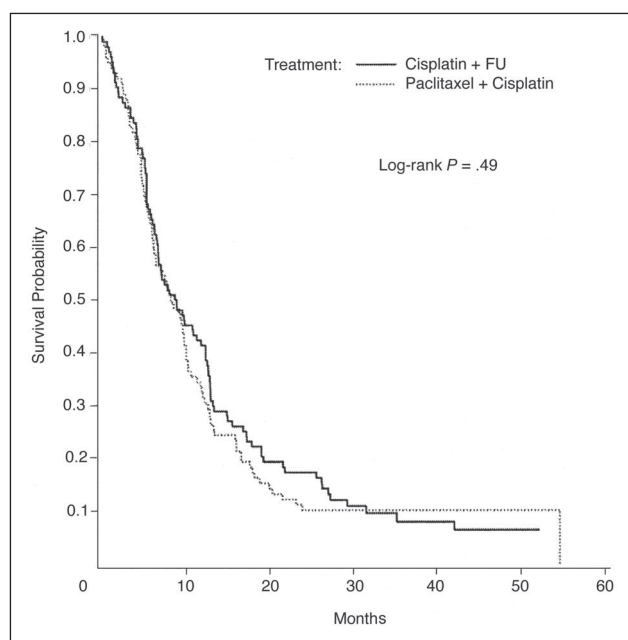


Fig 1. Kaplan-Meier plot for overall survival of eligible patients. One-year survival rates are as follows: cisplatin plus fluorouracil (5-FU), 41.4% (95% CI, 31.9% to 50.8%); cisplatin plus paclitaxel, 32.4% (95% CI, 23.1% to 41.6%); $P = .49$.

DISCUSSION

This randomized, phase III study aimed to compare the overall survival for a group of patients with advanced head and neck cancer treated with either infusional FU and cisplatin or paclitaxel (3-hour infusion) and cisplatin. No differences in this primary end point were detected. Furthermore, there were no major differences in the secondary end points of RR or toxicity.

Prior phase II studies of cisplatin and paclitaxel suggested a better overall survival and RR compared with cisplatin and FU. Historically, the median survival for infusional FU plus cisplatin was 4 to 6 months, with RRs in the 20% to 30% range.^{1,2,13} Phase II studies of the newer combination, paclitaxel (3-hour infusion) plus cisplatin, achieved slightly better median survivals of 10 to 11 months and RRs of 30% to 40%. While these phase II data suggested that CP might be better than CF, data from the current direct comparison do not support this conclusion.

Tumors of the head and neck may be difficult to measure reproducibly, especially when response is determined for assessable disease. To compensate for this possible confounder, the current study purposefully used overall survival and not RR as the primary end point. Nevertheless, only a small number of patients ($n = 17$) had assessable disease only, suggesting that in this analysis, both RR and survival are reliable outcomes. Potential explanations for the higher RR reported in earlier studies of the CP combination include inclusion of a greater proportion of patients

with assessable disease and less reliance on imaging in the prior studies. Both factors introduce measurement error, which may lead to overestimation of response.

Although survival did not differ, other aspects of each treatment combination such as toxicity and convenience might be considered when comparing the two. Infusional FU requires placement of an indwelling central venous catheter with the potential complications that include infection and thrombosis. Although these factors were not primarily evaluated in this trial, we noted only a slightly higher incidence of grade 3 infection in the CF group and no difference in thrombosis.

Although the Kruskal-Wallis test did not detect a significant difference in overall toxicity between the groups, subset analysis of toxicities may be of value for clinical decision making and practice. Stomatitis and diarrhea are common adverse effects of infusional FU and, as expected, were more common in the FU combination. Leukopenia, thrombocytopenia and anemia were also more common in the CF group, perhaps due to the higher dose of cisplatin administered with FU. Neurotoxicity, common to both cisplatin and paclitaxel, was equivalent.

Another measure of toxicity is the number of patients who switched from cisplatin to carboplatin. Of the 28 patients who switched to carboplatin, 21 were in the CF group. This may be due to the higher dose of cisplatin in the CF arm and may also account partially for the similar efficacy of each group given that carboplatin is less active than cisplatin in head and neck cancer. Because efficacy is no different and the associated toxicity may impact choice of therapy, an analysis of the quality of life and pain intensity measures was explored in a subset of patients in the trial and will be presented in a separate article.

Prior studies of the combination of paclitaxel and cisplatin employed a 24-hour infusion of paclitaxel. This schedule resulted in excessive hematologic toxicity, including febrile neutropenia and hospitalization in 30% to 40% of patients and several deaths attributed to infection. On the basis of these findings, the 24-hour schedule was not recommended for future use. This trial instead substituted a 3-hour infusion of paclitaxel on the basis of phase II studies that demonstrated similar efficacy but less toxicity compared with the 24-hour infusion. The efficacy and toxicity data from the current study support use of the 3-hour infusion. Although efficacy was preserved, severe hematologic toxicity was limited to febrile neutropenia in only 3% of patients receiving paclitaxel.

The combination of carboplatin and paclitaxel has gained favor for use in this population of patients, mostly due to the convenience and lower rate of renal toxicity of carboplatin compared with cisplatin. Recent phase II studies of carboplatin and paclitaxel demonstrated efficacy and toxicity results similar to the cisplatin-containing regimen used in this study.^{14,15} Because few patients in the current

study switched from cisplatin to carboplatin, the results presented here cannot be generalized to a carboplatin-containing combination.

While this study affirms the activity and utility of the combination of infusional FU and cisplatin, it fails to demonstrate superiority of the regimen containing paclitaxel. Despite active investigation of regimens containing newer cytotoxics, survival for patients with incurable squamous cell cancer of the head and neck remains poor. Few patients survive beyond 1 year. There is a pressing need

for drugs and combinations with greater efficacy, tolerability and convenience. Recently published experience with biologically targeted agents such as cetuximab, gefitinib, and erlotinib provide some hope that improvements are on the horizon.¹⁶⁻¹⁸

Authors' Disclosures of Potential Conflicts of Interest

The authors indicated no potential conflicts of interest.

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